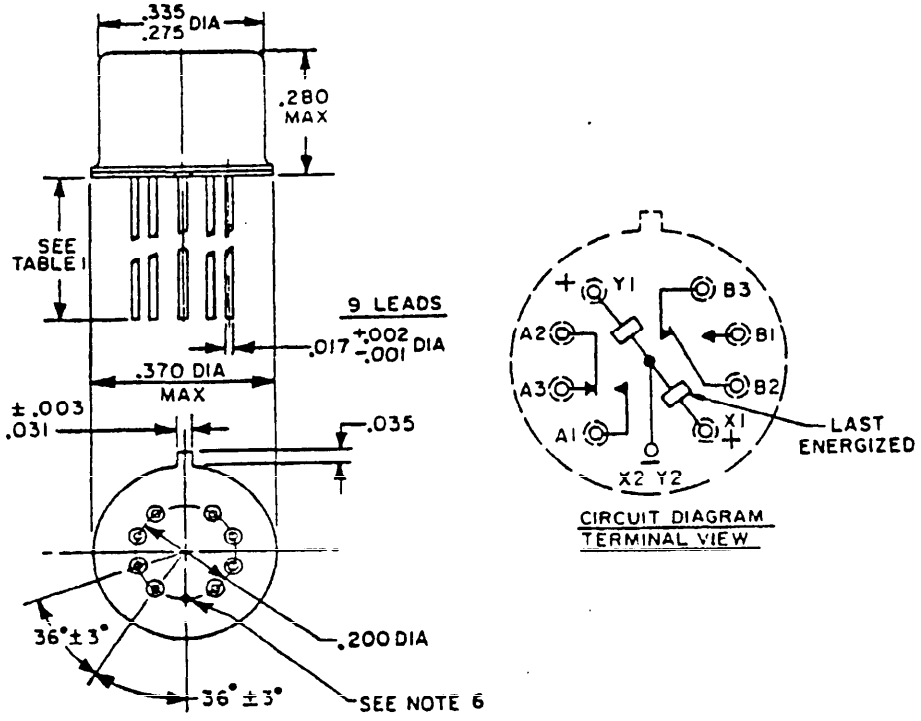
MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, DPDT, LOW LEVEL TO 1.0 AMPERE (LATCHING), INCLUDING INSPECTIONS OF APPENDIXES A AND B

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-39016.

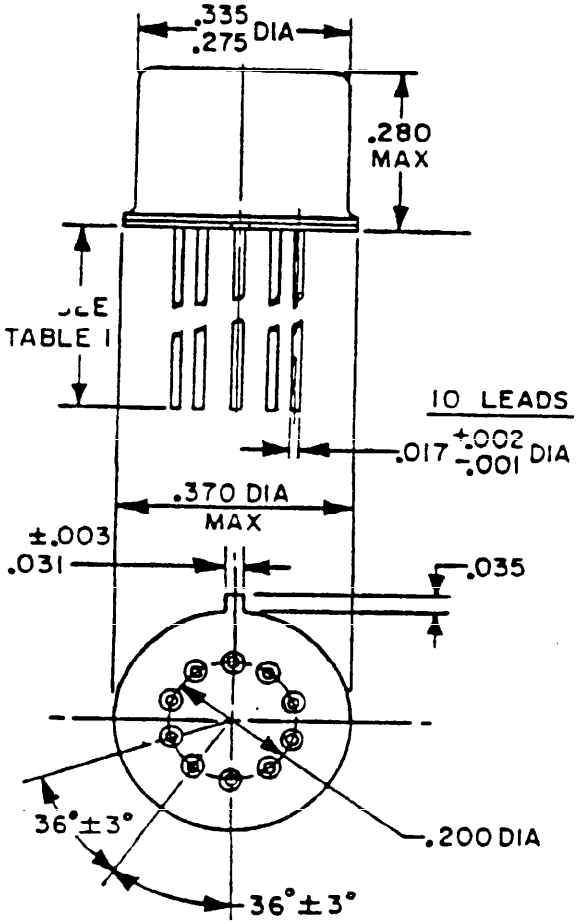


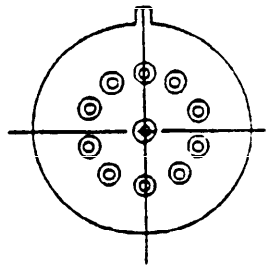
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers shown above for reference only. Numbers do not appear on the relay.
5. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
6. All leads shall be electrically insulated from the case, except for lead terminal -X2 -Y2, which is grounded to the case.
7. Circuit diagram shown on part is the terminal view.
8. Coil symbol optional in accordance with MIL-STD-1285.

FIGURE 1. Dimensions and configuration (round).

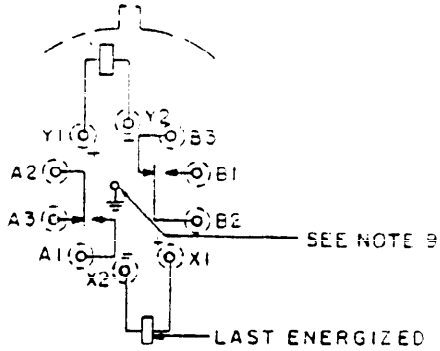
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TERMINAL VIEW
-025 THRU -030
(SEE NOTE 9)



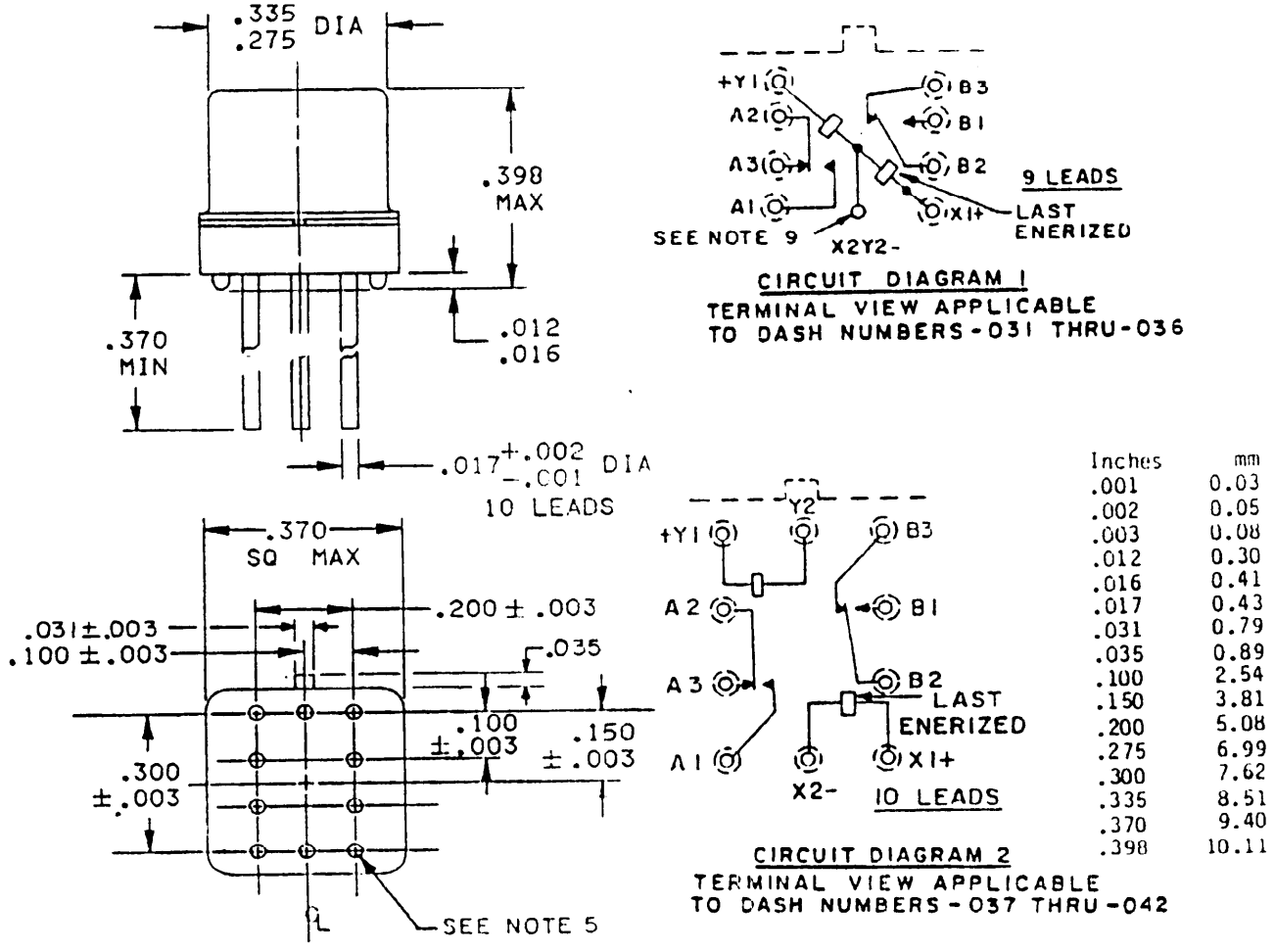
CIRCUIT DIAGRAM
TERMINAL VIEW

Inches	mm	Inches	mm
.001	0.03	.200	5.08
.002	0.05	.275	6.98
.003	0.08	.280	7.11
.017	0.43	.335	8.51
.031	0.79	.370	9.40
.035	0.89		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers shown above for reference only. Numbers do not appear on the relay.
5. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
6. All leads shall be electrically insulated from the case.
7. Coil symbol optional in accordance with MIL-STD-1285.
8. Circuit diagram shown on part is the terminal view.
9. M39016/612-025 through -030 shall be supplied with a case grounding pin welded to the relay header as shown.

FIGURE 2. Dimensions and configuration.

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Inches	mm
.001	0.03
.002	0.05
.003	0.08
.012	0.30
.016	0.41
.017	0.43
.031	0.79
.035	0.89
.100	2.54
.150	3.81
.200	5.08
.275	6.99
.300	7.62
.335	8.51
.370	9.40
.398	10.11

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Spreader pads shall be certified to MIL-M-38527, M38527/05-003, or M38527/05-013.
5. Dimensions are tolerance shown for the bottom view of the spreader pad are for the center to center locations of the holes in the spreader pad.
6. Shape optional within the envelope dimension.
7. Coil symbol optional in accordance with MIL-STD-1285.
8. Relays shall have a (+) sign placed on circuit diagram as shown.
9. All leads shall be electrically insulated from the case, except for lead terminal -X2 -Y2, circuit diagram I, which is grounded to the case.
10. Circuit diagram shown on part is the terminal view.
11. Terminal numbers shown above for reference only. Numbers do not appear on relay.

FIGURE 3. Dimensions and configuration relay with spreader pad attached.

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REQUIREMENTS:

Cleaning and small particle inspection: Applicable.

CONTACT DATA:

Load ratings:

High level (relay case grounded):

Resistive:

1.0 ampere at 28 V dc.

500 milliamperes at 115 V ac 400 Hz case not grounded.

250 milliamperes at 115 V ac 60 Hz case not grounded.

100 milliamperes at 115 V ac 60 and 400 Hz case grounded.

Inductive load: 0.2 ampere at 28 V dc wi 0.32 henry inductance.

Lamp: 0.10 ampere at 28 V dc.

Low level: 10 to 50 µA at 10 to 50 mV dc or peak ac.

Intermediate current: Applicable.

Contact resistance or voltage drop:

Initial: 0.125 ohm maximum (0.150 ohm maximum with spreader pad attached).

High level:

During life: Not more than 5 percent of open circuit voltage.

After life: 0.225 ohm maximum (0.250 ohm maximum with spreader pad attached).

Low level:

During life: 33 ohms maximum.

After life: 0.175 ohm maximum (0.200 ohm maximum with spreader pad attached).

Intermediate current:

During: 1 ohm maximum.

After: 0.225 ohm maximum (0.250 ohm maximum with spreader pad attached).

Contact bounce: 2 milliseconds (ms) maximum (applicable for after high level rated load life and intermediate current tests).

Contact stabilization time: 2.5 milliseconds maximum.

Overload (high level only): Two times rated current.

Neutral screen: Applicable.

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COIL DATA: See table I.

Operate time: 2.0 ms maximum over temperature range with rated coil voltage.

Release time: Not applicable.

ELECTRICAL DATA: 1/

Insulation resistance: 10,000 megohms minimum at 500 V dc, except the resistance between coil and case at high temperature shall be 1,000 megohms minimum.

Dielectric withstanding voltage:

	Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure and all contacts both in the energized and deenergized positions	500	125 All terminals to case
Between case, frame, or enclosure and coil(s)	[illegible]	
Between all contacts and coil(s)	500	
Between open contacts in the energized and deenergized positions	500	
Between contact poles	500	
Between coils of dual coil relays (figure 2 and circuit diagram 2 of figure 3)	500	

ENVIRONMENTAL DATA:

Temperature range: -65°C to +125°C.

Vibration (sinusoidal): MIL-STD-202, method 204. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts.

Vibration (random): MIL-STD-202, method 214, test condition IG. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts. Applicable to qualification, group C, and group A inspection testing.

Shock (specified pulse): MIL-STD-202, method 213, test condition B (75 g's). Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts.

Magnetic interference: Applicable.

Resistance to soldering heat: Applicable.

Acceleration: Applicable.

Salt atmosphere (corrosion): In accordance with method 1041, MIL-STD-750.

Particle impact noise detection: Applicable.

1/ Dielectric withstanding voltage and insulation resistance are not applicable between coil and case or from coil to coil on figure 1 relays and circuit diagram 1 relays on figure 3.

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Part or Identifying Number (PIN): M39016/612- (dash number from table I and suffix letter designating failure rate level).

TABLE I. Dash numbers and characteristics. 1/ 2/

Dash number 3/				Coil voltage (V dc) 4/		At 25°C		At 125°C
Lead length .187 ±.010	Lead length .500 min	Figure	Spreader pad 5/ Figure 3	Rated	Max	Coil resistance ohms ±10%	Specified pickup (latch/ reset) value (voltage) (V dc)	Specified pickup (latch/ reset) value (voltage) (V dc)
001	013	1	031	5.0	6.0	61	2.8	3.5
002	014	1	032	6.0	8.0	120	3.5	4.5
003	015	1	033	9.0	12	280	5.3	6.8
004	016	1	034	12	16	500	7.0	9.0
005	017	1	035	18	24	1,130	10.5	13.5
006	018	1	036	26.5	32	2,000	14.2	18.0
007	019	2	037	5.0	6.0	61	2.8	3.5
008	020	2	038	6.0	8.0	120	3.5	4.5
009	021	2	039	9.0	12	280	5.3	6.8
010	022	2	040	12	16	500	7.0	9.0
011	023	2	041	18	24	1,130	10.5	13.5
012	024	2	042	26.5	32	2,000	14.2	18.0
---	025 6/	2	---	5.0	6.0	61	2.8	3.5
---	026 6/	2	---	6.0	8.0	120	3.5	4.5
---	027 6/	2	---	9.0	12	280	5.3	6.8
---	028 6/	2	---	12	16	500	7.0	9.0
---	029 6/	2	---	18	24	1,130	10.5	13.5
---	030 6/	2	---	26.5	32	2,000	14.2	18.0

1/ Warning: When latching relays are installed in equipment, the latch and reset coils should not be pulsed simultaneously. Coils should not be pulsed with less than the nominal coil voltage and the pulse width should be a minimum of three times the specified operate time of the relay. If these conditions are not followed, it is possible for the relay to be in the magnetically neutral position.

2/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits not recommended for subsequent use in low level applications.

3/ The suffix letter M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles): M, 1.0; P, 0.1; R, 0.01. Example, 025M - - - - - 030R.

4/ Caution: The use of any coil voltage less than the rated voltage will compromise the operation of the relay.

5/ Relays supplied with spreader pads (-031 through -042) shall have the pad rigidly attached.

6/ Relays are supplied with a case grounding pin welded to the header (see figure 2).

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PHYSICAL DATA:

Terminal strength (method 211, MIL-STD-202):

Pull test: Test condition A, 1 pound pull.

Bend test: Test condition C, 1/2 pound load.

Twist test: As specified in MIL-R-39016.

Solderability: Applicable.

Dimensions and configurations: See figures 1, 2, and 3.

Weight: 2.84 grams (0.10 ounce) maximum, 3.09 grams (0.109 ounce) maximum with spreader pad attached.

Seal: Hermetic, adjunct sealant shall not be used.

Minimum marking: Military PIN, "J" with the date code (example J8830), circuit diagram, manufacturer's name or source code.

LIFE TEST REQUIREMENTS:

High level: 100,000 cycles per relay.

Low level: 100,000 cycles plus 900,000 cycles mechanical life.

QUALIFICATION INSPECTION:

Qualification inspection and sample size: See table II.

Failure rate qualification: The manufacturer shall qualify his products to failure rate level "M" initially.

TABLE II. Qualification inspection and sample size. 1/

Single submission	Group submission	
33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable	M39016/612-018	33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable
	M39016/612-013 M39016/612-014 M39016/612-015 M39016/612-016 M39016/612-017 M39016/612-019 M39016/612-020 M39016/612-021 M39016/612-022 M39016/612-023 M39016/612-024	2 units each PIN Qualification inspection, group I and group II
	M39016/612-030	1 unit terminal strength and solderability
	M39016/612-036	2 units each PIN 3/

See footnotes on next page.

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1/ For retention of qualification or extension of qualification to lower failure rate levels, all life test data accumulated on (or specified in) MIL-R-39016/12, MIL-R-39016/629, and MIL-R-39016/630 may be used in addition to MIL-R-39016/612. Prior to performance of retention of qualification testing, the relay manufacturer shall preselect the sampling plan.

2/ The number of units required for qualification testing shall be increased as required in group V, table II, MIL-R-39016, if the contractor elects to test the number of units permitting one or more failures. Prior to performance of qualification testing, the relay manufacturer shall preselect the sampling plan.

3/ Qualification inspection of relays supplied with spreader pads shall be tested as specified in the qualification inspection table of MIL-R-39016 in the order shown below:

Before installation of the pad:

Vibration (sinusoidal) test duration shall be 10 minutes, vibration (random), particle impact noise detection, screening.

After installation of the pad:

Insulation resistance, dielectric withstanding voltage, static contact resistance, specified pickup, hold, and dropout values (voltages), coil resistance, operate and release time, contact dynamic characteristics, coil transient suppression (when specified), solderability, seal, visual and mechanical inspection (external).

Group A inspection: Group A inspection for relays supplied with spreader pads shall be as specified below:

Before installation of the pad: Perform subgroups 1 and 2 of the group A inspection table.

After installation of the pad: Perform subgroups 3 and 4 of the group A inspection table.

Qualification inspection (reduced testing) and sample size: If the relays produced for MIL-R-39016/612 are similar in construction and design, except for the additional inspections specified herein, to the relays produced for MIL-R-39016/12, then reduced testing may be performed concurrent with or subsequent to successful qualification of MIL-R-39016/12 at the "M" failure rate level. See table III.

TABLE III. Qualification inspection (reduced testing).

Examination or test
2 units each coil voltage, group A inspection table. 1 unsealed sample unit for internal examination.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - AS, OS, SH
Air Force - 99
DLA - ES
NSA - NS

User activities:
Navy - MC
Air Force - 11, 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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